

RESEARCH LETTER

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Uplift of the ionosphere by tropospheric and thermospheric Lamb waves induced by the 2022 Tonga volcanic eruption

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Abstract

On 15 January 2022, the Tonga submarine volcano eruption generated atmospheric shocks, air pressure perturbations, tsunamis, etc., inducing traveling atmospheric disturbances (TADs) and traveling ionospheric disturbances (TIDs). Data from the Himawari-8 satellite, ionosondes, as well as nearby barometers and tide gauges are employed to study the perturbations over Taiwan and Japan. Ionograms reveal that TIDs associated with the thermospheric and tropospheric Lamb wavefronts propagate with the northwestward (upward) speeds of 539.0 ± 22.3 and 338.3 ± 4.0 m/s (21.0 ± 5.0 and 12.4 ± 3.3 m/s), which further uplift the ionosphere by 86–155 and 50–110 km, respectively. The latter TID is followed by distinct TAD signatures, including prominent near-Earth surface air pressure pulses and significant sea level fluctuations, whereas the former one is not. The two TIDs significantly uplift the ionosphere which increase the growth rate of Rayleigh–Taylor instability, and result in the occurrence of ionospheric spread-F layers.

Keywords Lamb wave, Ionosphere, Tonga volcano, Ionosonde

Introduction

At about 04:10 UT on 15 January 2022, the Tonga submarine volcano (20.536°S, 175.382°W) erupted violently with a volcanic explosivity index (VEI) of 5–6 (Yuen et al. 2022), which produced a significant plume height, reaching over 50 km into the lower mesosphere (Carr

et al. 2022; Yuen et al. 2022). Meteorological satellite images and barometer records show traveling atmospheric disturbances (TADs) in the tropospheric Lamb mode with the horizontal speed of about 315 m/s globally (Amores et al. 2022; Matoza et al. 2022; Wright et al. 2022; Liu et al. 2023). Tsunami waves associated with pressure perturbations induced by the Tonga eruption were found beyond the Pacific Ocean coastline (Carvajal et al. 2022; Lynett et al. 2022; Omira et al. 2022). On the other hand, the TADs further propagated into the ionosphere, becoming traveling ionospheric disturbances (TIDs) that were detected by the total electron content (TEC) derived from ground-based Global Navigation Satellite System (GNSS) receivers (Astafyeva et al. 2022; Lin et al. 2022; Themens et al. 2022; Wright et al. 2022; Zhang et al. 2022; Heki 2022).

On the other hand, Shinbori et al. (2023) observed electron density perturbations recorded by two ionosondes, located at Guam and Wake, approximately 4850–5780 km from the Tonga volcano, propagating at

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horizontal speeds of about 480–540 m/s, implying that acoustic waves propagate in the thermosphere. Liu et al. (2023, 2024) examined the magnetic field changes of 69 INTERMAGNET magnetometers and the ion density measured by the FORMOSAT-7/COSMIC-2 satellite, and found that both tropospheric and thermospheric Lamb wavefronts with horizontal speeds of 315 and 487 m/s, respectively, efficiently trigger TIDs and modify ionospheric currents on a global scale.

In this study, we investigate TIDs in ionograms related to thermospheric and tropospheric Lamb wavefronts with reference horizontal speeds of 315 and 487 m/s in Taiwan and Japan, respectively. Near each ionosonde, the Earth's surface air pressure monitored by a barometer and the sea level recorded by a tide gauge are also used to study TADs underneath the TID. This allows us studying the impact of the thermospheric and tropospheric Lamb waves across various atmospheric layers approximately 7800–8600 km from the Tonga volcano.

Observation and result

Full-disk images from the Japan Meteorological Agency (JMA)'s Himawari-8 satellite (Bessho et al. 2016) are used to observe TADs in upper-level tropospheric water vapor at about 8.2 km altitude, with a wavelength of 6.2 μm in the infrared band (#8), as suggested by Wimmers et al. (2018) and Otsuka (2022). The images are processed using the second-time derivative technique (Otsuka 2022). Complementary to this, five ionosondes, along with their nearby barometers and tide gauges over Taiwan and Japan, are employed to investigate the TADs and TIDs induced by the Tonga volcano eruption (see also Table S1 in Supporting Information). The ionograms with 6-min intervals in Taiwan and 5-min intervals in Japan are compressed for analysis (Igi 1992). In Taiwan, the surface air pressure and sea level are recorded every minute, while in Japan, the time resolutions of air pressure data and sea level records are 10 min and 1 min, respectively. Here, a one-hour high-pass filter is applied to sea level variations for observational clarity.

Figures 1 and 2 illustrate that the thermospheric Lamb wavefront with a horizontal speed of 487 m/s reached Taiwan at 09:02 UT and the tropospheric Lamb wavefront with a speed of 315 m/s arrived in Taiwan at 11:40 UT. Figure 1 also depicts locations of ionosondes, barometers, and tide gauges in Taiwan and Japan. Figure 2a adapted from Liu et al. (2023) displays deviations in the northward (ΔB_x), eastward (ΔB_y), and azimuth of horizontal (ΔH) component, while Fig. 2b depicts ionograms every 30 min in Taiwan. Figure 2a shows that the minimum of ΔB_x precedes the thermospheric Lamb wavefront by about 27 min, while the central time point of the ΔH azimuth directed toward the Tonga volcano precedes

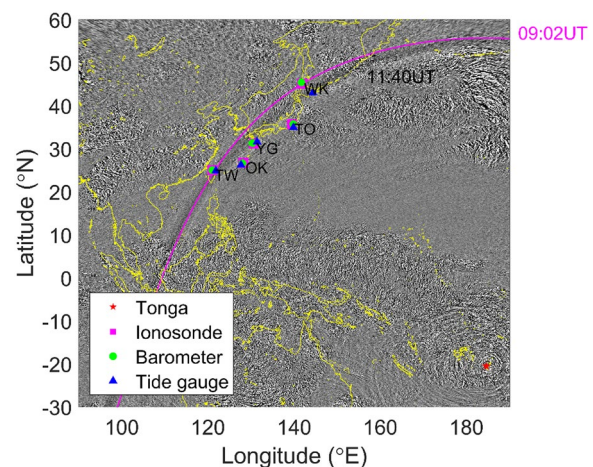
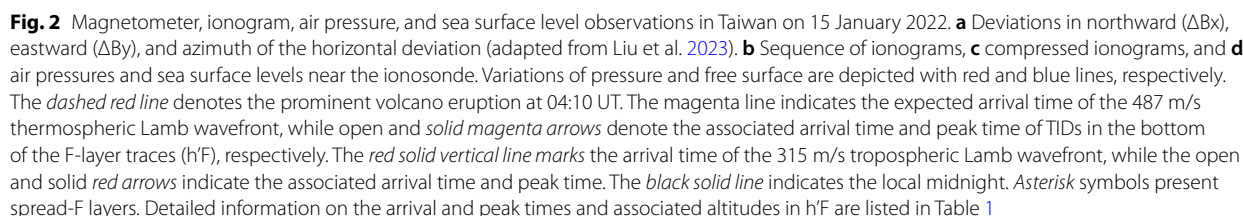


Fig. 1 Differential image from the Himawari-8 satellite showing the tropospheric Lamb wavefront at 11:40 UT, along with the thermospheric Lamb wavefront (magenta curve) at 09:02 UT, the location of the Tonga volcano (red star), and observation sites, including barometers (green dots), tide gauges (blue triangles), and ionosondes (magenta cubes) in Taiwan and Japan, which are located approximately 7800–8600 km from the Tonga volcano, on 15 January 2022

the tropospheric Lamb wavefront by about 20 min. Figure 2b illustrates that after the passages of the ΔB_x and ΔH perturbations and the two Lamb wavefronts, the ionograms ascend. Following the two ascendances, the range spread-F layers start appearing at 12:30 UT (20:30 LT).

Figures 2c–d and 3 show responses of compressed rapid-run ionograms, air pressures, and filtered sea levels to the thermospheric and tropospheric Lamb wavefront passages in Taiwan and Japan. In Taiwan (Fig. 2c–d; see also Movie S1 in Supporting Information S1), the bottom of the F-layer traces ($h'F$) reaches a local minimum of 215 km before starting to ascend at 08:48 UT (16:48 LT, UT+8, denoted as t_a), about 14 min before the thermospheric Lamb wavefront arrival at 09:02 UT (17:02 LT). The $h'F$ peaks at 311 km at 10:06 UT (18:06 LT, t_p), descends to 228 km at 11:06 UT (19:06 LT), and oscillates around 220–227 km between 11:12 and 11:42 UT (19:12–42 LT). The $h'F$ ascent of 96 ($\Delta h'F = 311 - 215$) km occurred within $\Delta t = 78$ min (t_a to t_p), corresponding to an average upward speed ($\Delta h'F / \Delta t$) of about 20.5 m/s. In Japan (Fig. 3a, c, e, g; see also Movies S2–S5 in Supporting Information S1), observations at WK, YG, OK, and TO show similar trends. The $h'F$ s, initially at 179–197 km, began ascending at 08:00–08:45 UT (17:00–17:45 LT, UT+9), and reached peaks of 283–334 km at 09:40–10:15 UT (18:40–19:15 LT), which corresponds to an ascent of 86–155 km within 90–105 min. These upward motions started 15–39 min before the thermospheric Lamb



Following the thermospheric wavefront passage, in Taiwan, prior to the arrival of the tropospheric Lamb wavefront at 11:42 UT (19:42 LT), h'F at 225 km altitude

Ionosonde	Location (N, E)	Distance (km)	Thermospheric Lamb Wave						Tropospheric Lamb Wave					
			487 m/s Frnt time (UT)	Arrival/Peak time (UT)	Horizontal speed (m/s)	Arrival/Peak Height (km)	Δh°F (km)	Vertical Speed (m/s)	315 m/s Frnt time (UT)	Arrival/Peak time (UT)	Horizontal speed (m/s)	Arrival/Peak height (km)	Δh°F (km)	Vertical Speed (m/s)
Wakkanai (WK)	(45.2, 141.8)	8493	0900	0845 / 1015	514.7	197/283	86	15.9	1139	1100 / 1245	345.3	170/280	110	17.5
Kokubunji (TO)	(35.7, 139.5)	7850	0839	0800 / 0940	568.8	180/304	124	20.7	1105	1040 / 1230	335.5	170/230	60	9.1
Yamagawa (YG)	(31.2, 130.6)	8139	0849	0815 / 0940	553.6	179/334	155	30.4	1121	1055 / 1235	334.9	180/230	50	8.3
Okinawa (OK)	(26.7, 128.2)	8030	0845	0815 / 1000	546.3	187/298	111	17.6	1115	1043 / 1220	340.6	170/250	80	13.7
Singwu (TW)	(25.0, 121.0)	8535	0902	0848 / 1006	511.7	215/311	96	20.5	1142	1114 / 1248	335.5	225/300	75	13.3
					Average h°F horizontal speed: 539.0±22.3 m/s		Average h°F vertical speed: 21.0±5.0 m/s				Average h°F horizontal speed: 338.3±4.0 m/s		Average h°F vertical speed: 12.4±3.3 m/s	
Air pressure	Location (N, E)	Distance (km)	487 m/s Frnt time (UT)						315 m/s Frnt time (UT)	Peak time (UT)	Horizontal Speed (m/s)	ΔPressure (hPa)		
WAKKANAI	(45.4, 141.7)	8528	0901						1140	1210	295.6	1.8		
TOKYO	(35.7, 139.8)	7828	0838						1104	1130	296.6	1.7		
MAKURAZAKI	(31.3, 130.3)	8145	0849						1122	1140	302.5	1.1		
NAGO	(26.6, 128.0)	8011	0845						1115	1130	304.6	0.9		
NCU	(25.0, 121.0)	8494	0902						1141	1157	304.6	1.8		
											Average speed: 300.8±3.9 m/s			

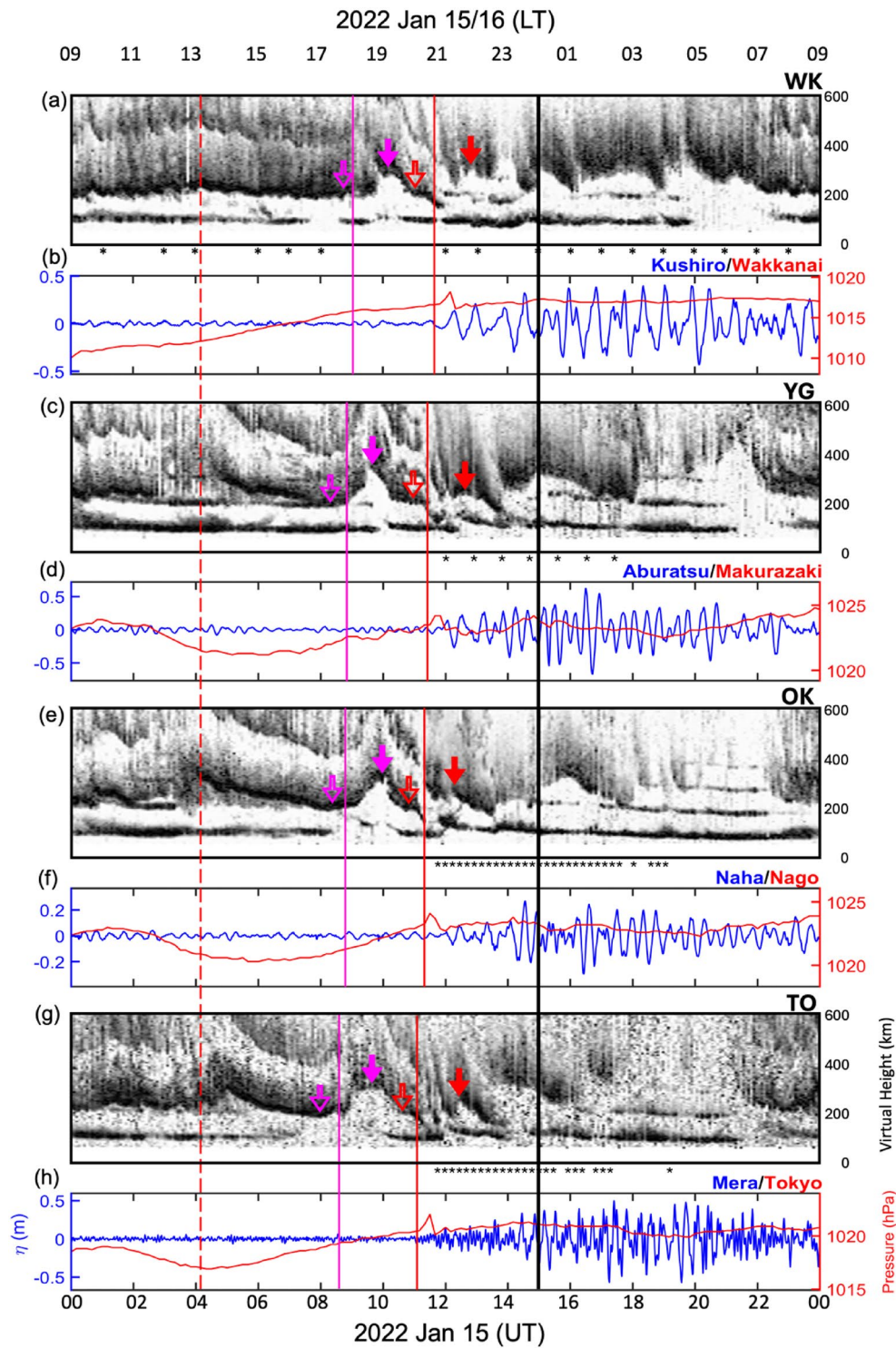


Fig. 3 Observations of ionograms (a, c, e, g) and air pressures and sea surface levels (b, d, f, h) at four stations in Japan on 15 January 2022. Symbols and lines are the same as in Fig. 2. Asterisks below the ionograms (a, c, e, g) are denoted regarding the f-PLOT of the National Institute of Information and Communications Technology (<https://wdc.nict.go.jp/ionosphere/en/archive/>, last access: 2024/04/09)

started ascending at 11:14 UT (19:14 LT), reaching a peak of 300 km at 12:48 UT (20:48 LT). This 75 km ascent occurred over 94 min, corresponding to an average upward speed of 13.3 m/s. In Japan, h'Fs began ascending 25–39 min before the tropospheric Lamb wavefront arrival, rising by 50–110 km within 97–110 min. The average upward speed across the five stations is 12.4 ± 3.3 m/s (Table 1). Note that significant range-spread-F layers are observed at TW in Taiwan during 12:30–20:00 UT (20:30–04:00 LT), and at stations in Japan during 11:45–19:45 UT (20:45–04:45 LT) after the thermospheric and tropospheric Lamb wavefronts pass, except with an earlier and extended occurrence time at the high-middle station of WK (see also Supporting Information, Figure S1). Note that the ionosphere has been uplifted at rather high altitude after the thermospheric and tropospheric Lamb wavefronts pass.

Figures 2 and 3 further display that the air pressures yield a sudden pulse after gradually increasing, while the sea levels fluctuate significantly as the tropospheric Lamb wavefront arrives in Taiwan and Japan. In contrast, variations in the air pressure and sea level upon the thermospheric Lamb wavefront arrival are much less pronounced. As the tropospheric Lamb wavefront arrives in Taiwan and Japan, the air pressures yield a sudden pulse with the maximums of $\Delta P = 0.9$ – 1.8 hPa within about 15–30 min, followed by the sea level fluctuations of approximately 7–27 cm. These TAD signatures generally agree with the results reported by Ren et al. (2023). After the TAD features, prominent sea level fluctuations reached a maximum height of 0.43 m at the Wushi station in Taiwan at 15:50 UT and 0.61 m at Aburatsu in Japan at 16:29 UT, which concurs with the arrival time of conventional tsunami waves traveling with the shallow water speed, $c_{LW} = \sqrt{gh}$, where g represents the acceleration of gravity and h stands for the ocean water depth.

Discussion

Table 1 presents the location of ionosondes and barometers; the distance of each observation site from the volcano; the arrival times of two Lamb wavefronts; and the arrival/peak times (t_a/t_p), horizontal speeds, heights (arrival/peak and uplifted heights, $\Delta h'$), vertical speeds, and pressure variations of TIDs and TADs observed at each station. With the obtained dataset, the horizontal speeds of TADs and TIDs can be calculated by dividing the distances by the travel times (arrival time minus the volcano eruption time). These individual speeds are then averaged to determine the mean horizontal speeds. Figure 4a and Table 1 depict that based on the arrival time of t_a in h'F, the overall mean horizontal TID speeds associated with the thermospheric and tropospheric Lamb wavefronts are about 539.0 ± 22.3 and 338.3 ± 4.0 m/s,

respectively. The associated TIDs traveling faster than the Lamb wavefronts suggest that the bow waves have been formed. On the other hand, based on the peak times of the TADs recorded by the five barometers, the average horizontal speed of the surface air pressures is about 300.8 ± 3.9 m/s over Japan and Taiwan.

To better understand the TADs and TIDs, the US Naval Research Laboratory Mass Spectrometer and Incoherent Scatter Radar Extended (NRLMSISE-00) model (Picone et al. 2002) is performed to obtain the parameters necessary for deriving the sound speed profile (Shinbori et al. 2023). Additionally, to find the vertical interaction among the Earth's surface atmosphere, troposphere, and thermosphere, the neutral mass density is used to calculate the amplification factor of disturbances in various altitudes. Assuming energy conservation as TADs propagate from the troposphere into the thermosphere/ionosphere and become TIDs, the amplitude of TADs/TIDs is expected to be inversely proportional to the square root of the neutral density (Liu et al. 2012). The simulation is configured with a time setting of 12:00 UT on January 15, 2022, which is close to the arrival time of the tropospheric Lamb wavefront in Taiwan, at the geographic coordinates of 25°N latitude and 121°E longitude.

We then plot the mean horizontal speeds with the instruments' observing altitudes along the sound speed profile derived from the NRLMSISE-00 model. Figure 4b illustrates that the TIDs in h'F at 183–192 km altitude associated with the thermospheric and tropospheric Lamb wavefronts propagate with the speed of 539.0 ± 22.3 and 338.3 ± 4.0 m/s, respectively. The horizontal speeds of thermospheric Lamb wave TIDs are close to the acoustic waves at the bottom side of F-region at about 120–130 km altitude. This observation is consistent with the findings of Shinbori et al. (2023), which reported traveling speeds of about 480–540 m/s at two ionosonde stations in Guam and at Wake, where are approximately 4850–5780 km from the Tonga volcano. Meanwhile, the horizontal speeds of tropospheric Lamb wave TIDs are nearly identical to the observations of the upper-level troposphere (Fig. 1).

Furthermore, Figs. 2–4 show that TIDs in h'Fs lead the tropospheric and thermospheric Lamb wavefronts by about 25–39 and 14–39 min, respectively. Liu et al. (2023) reported that TIDs of changes in the northward magnetic fields lead the thermospheric Lamb wavefront by about 12–30 min, while those in the horizontal magnetic field lead the tropospheric wavefront by about 30–40 min (also see Fig. 2a, b). The agreements between the current study and Liu et al. (2023) again show that both tropospheric and thermospheric Lamb wavefronts can trigger bow waves, which could be developed as the disturbance travels as fast or faster than the ambient gravity or sound

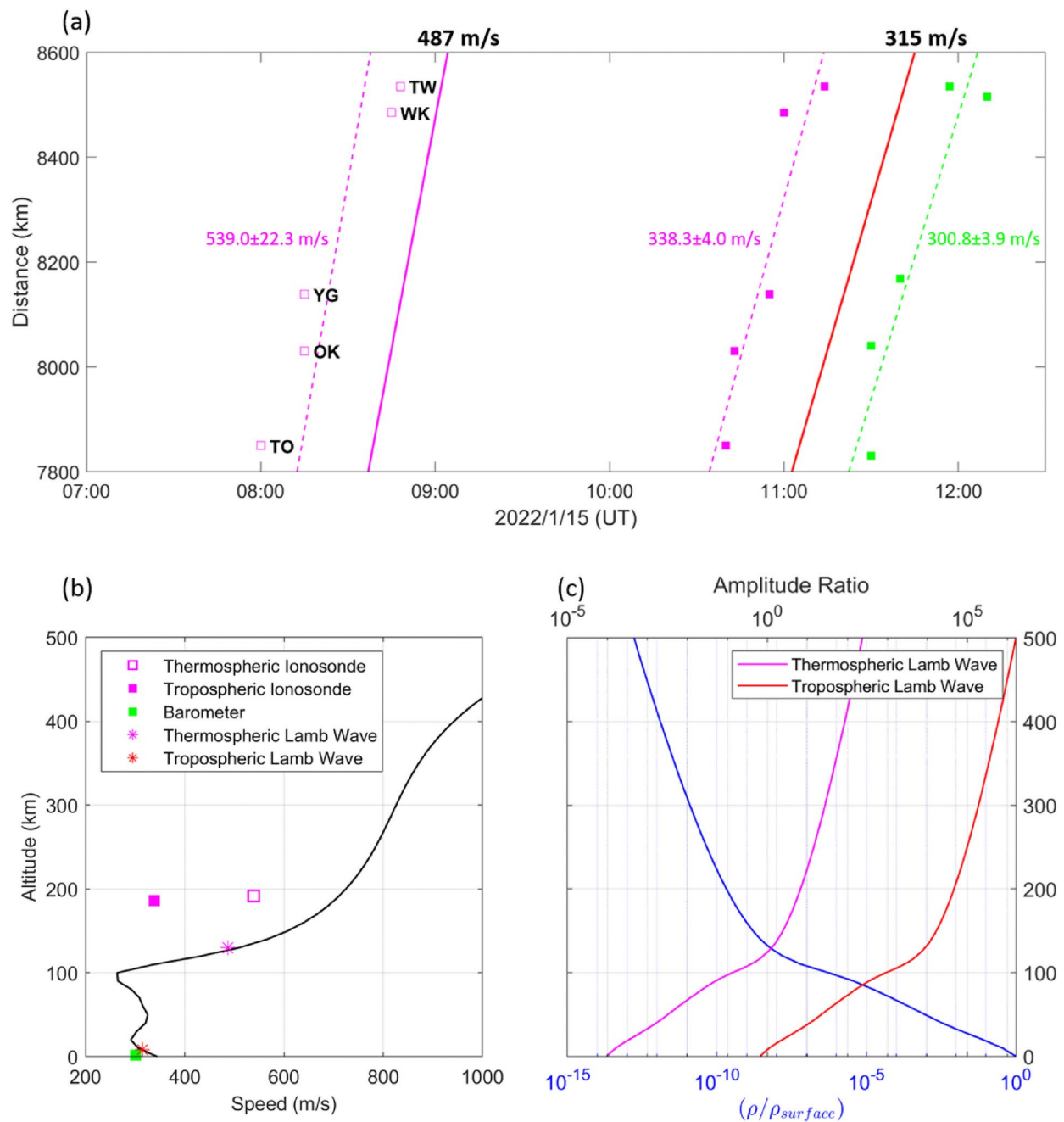


Fig. 4 Horizontal speeds and altitudinal profiles of observed TIDs and TADs associated with Lamb wavefronts. **a** Arrival times versus distances to the Tonga volcano of thermospheric and tropospheric Lamb wavefronts with reference speeds of 487 and 315 m/s, respectively, as well as the associated signatures of TADs in surface pressures and TIDs in ionograms at each observation site. Numbers are the averaged horizontal speeds. **b** Comparison of the averaged horizontal speeds of each observation, the thermospheric and tropospheric Lamb modes, and the atmosphere sound speed profile. Panels **a** and **b** utilize the common legend for consistency. **c** Profiles of the neutral mass density ratio and the amplification factor of the thermospheric and tropospheric Lamb modes derived from NRLMSISE-00

wave speeds (Chimonas 1970; Zhang et al. 2017; Lin et al. 2018; Sun et al. 2018).

Utilizing multiple sources and types of data at different altitudes and locations, we found that the Lamb waves

travel at ~8 km altitude in the upper-level troposphere and at 120–130 km altitude in the lower thermosphere. Figures 2 and 3 show that the tropospheric Lamb waves can trigger prominent TADs in the surface air pressures

and sea levels, while the thermospheric Lamb waves could not. The amplification factors can be calculated to understand the vertical influence of these waves (Liu et al. 2012). Let the neutral mass density computed by the NRLMSISE-00 model on the ground be 1, Fig. 4c depicts the ratios of the density at the surface to that at the upper tropospheric height of 8 km, the Pedersen/Hall conductivity peak height of 100 km, the low thermospheric height of 120–130 km, and the base height of the F-region of 200 km are about 0.45, 5.2×10^{-7} , 1.1×10^{-8} , and 2.5×10^{-10} , respectively. Therefore, the amplitude ratios of tropospheric Lamb waves at 8 km altitude to those of TADs/TIDs at the Earth's surface, 100, 120, and 200 altitudes are 0.67, 926.55, 6.9×10^3 , and 4.6×10^4 , respectively. These ratios show that tropospheric Lamb waves can prominently trigger TADs, influencing surface pressures and sea levels, and significantly disturb TIDs of h'F in the ionosphere. In contrast, the amplitude ratios of thermospheric Lamb waves at 120–130 km altitude to those of TADs/TIDs at the Earth's surface, 100, and 200 km altitudes are 9.7×10^{-5} , 0.13, and 6.75, respectively. This suggests that while thermospheric Lamb waves can disturb h'F in the ionosphere, no significant changes in the surface air pressures and sea levels are observed upon their arrival due to the small amplitude ratio of 9.7×10^{-5} . It is interesting to find that the thermospheric Lamb TIDs horizontally traveling close to the acoustic waves of 539.0 ± 22.3 m/s at 120–130 km altitude is coincidentally around the changing point of the ratio of the neutral density (Fig. 4c). Note that the tropospheric Lamb TADs in Himawari-8 images at ~8 km altitude is right below the coldest temperature at the tropopause, where yields the greatest temperature gradient.

Upon the arrival of the thermospheric and tropospheric Lamb wavefronts, the ionosphere can be uplifted with the average speeds of 21.0 ± 5.0 and 12.4 ± 3.3 m/s, respectively. The ionospheric signatures observed on the event day exhibit pronounced uplifts and temporal variations that are not present on magnetically quiet reference days (see also Text S1 and Figures S1–2). It is known that equatorward and poleward neutral winds can, respectively, move the F-region plasma to higher and lower altitudes along the geomagnetic field lines due to the ion-neutral collision. Additionally, eastward and westward electric fields can raise and lower the ionosphere as well (Davies 1990; Kelley 2009). Harding et al. (2022) and Le et al. (2022) used ICON-MIGHTI observations, showing that the neutral winds associated with tropospheric Lamb waves triggered by the January 2022 Tonga volcanic eruption are about 100–200 m/s. Therefore, the northwestward neutral winds caused by the volcanic eruption would result in downward vertical speeds in Taiwan and Japan; the direction of movement is opposite to the present study's observation (Figs. 2,

3; Table 1). On the other hand, Liu et al. (2023) and Fig. 2a show that the eastward component of the dynamo-generated electric fields and the northward component of the Earth's magnetic field results in prominent upward drifts $\mathbf{V} = \mathbf{E} \times \mathbf{B}/B^2$ drift. Using the total and horizontal components of the Earth's magnetic field from the International Reference Magnetic Field (IGRF), which are 41,065.6 and 32,677.6 nT in Taiwan, along with variables in Table 1, the dynamo electric fields associated with the thermospheric and tropospheric Lamb wavefronts are calculated to be 1.1 and 0.6 mV/m in the eastward direction, respectively. These values are comparable to the daily dynamo electric fields observed in the low-latitude ionosphere (Woodman 1970; see also Supporting Information, Figure S2). Meanwhile, reduced E-region conductivity by the passage of Lamb wave after the Tonga volcano eruption during sunset enables strong pre-reversal enhancement (PRE) in the January solstice period, amplified by enhanced zonal winds and negative storm response over equatorial ionization anomaly (Rajesh et al. 2022). In short, due to the strong eastward dynamo/PRE electric fields, the north-upward $\mathbf{E} \times \mathbf{B}/B^2$ drift overpowers the north-downward owing to the neutral winds, which results in the significant uplift of the ionosphere after the arrival of the thermospheric and tropospheric Lamb wavefronts.

While Shinbori et al. (2023) observed a rapid increase of h'F with a magnitude of 115–155 km but did not interpret the feature later due to the occurrence of spread-F, Liu et al. (2023) noted that sporadic E (Es) layers and spread-F layers began remarkably ascending upon the arrival of the tropospheric Lamb wavefront, as observed with the ionosonde in Taiwan. With satellite and ionosonde observations, Sun et al. (2022) report that TIDs induced by the Tonga volcanic eruption arrived in the Asian sector before sunset, elevating the F-layer (09:30–11:30 UT), which seeded Rayleigh–Taylor instability and enhanced the growth of equatorial/low-latitude plasma bubbles. Figure 2 depicts that after prominent ionospheric uplifts by TIDs of the thermospheric and tropospheric Lamb waves, significant range spread-F layers appear in ionograms observed by ionosondes at TW in Taiwan as well as OK and TO in Japan. The two horizontal speeds 539.0 ± 22.3 and 338.3 ± 4.0 m/s and the two peak times of t_p at 10:00 and 12:30 UT in Table 1 show that the wavelength of TIDs triggered by thermospheric and tropospheric Lamb waves are of about 3000 km.

Based on Kelley (2009), the generalized Eq. (1) of the growth rate of Rayleigh–Taylor instability can be expressed as follows:

$$\gamma = \frac{g}{Lv_{in}} + \frac{E}{LB}, \quad (1)$$

where g , L , ν_{in} , E , and B denote the acceleration of gravity, the gradient scale length of the plasma density, the ion-neutral collision frequency, the electric field, and the Earth's magnetic field, respectively. A feature predicted by Eq. (1) is a height dependency for γ due to the collision frequency term in the growth rate denominator and the eastward electric field; the higher the layer, the smaller ν_{in} and the larger the instability growth rate (Kelley 2009).

Shinbori et al. (2023) observe increased rate of TEC index (ROTI) values and uplift of the F-region prior to the arrival of air pressure waves, displaying the link between atmospheric disturbances and the formation of equatorial/low-latitude plasma bubbles. Their findings suggest that the TIDs propagating in the thermosphere had triggered the ionosphere's vertical motions, enhancing the Rayleigh–Taylor instability's growth rate. On the other hand, Rajesh et al. (2022) report that the passage of Lamb waves through the E-region results in reduced conductivity, which in turn produces intense electric fields. Under such conditions, rapidly up-drifting irregularities are generated with seed perturbations from Lamb wave-driven gravity waves, producing super plasma bubbles that reach extremely poleward latitudes (Chou et al. 2021). TIDs driven by gravity waves can have a positive or negative impact on bubble growth, depending on the positive/negative phases of the wavefronts of TIDs. For this reason, the wavelength of the gravity waves driven by the Tonga-triggered Lamb wave is crucial. If the wavelength is long enough, such as 3000 km in Figs. 2 and 3, the coincidence of positive phase can be effective for a sufficiently long time to continuously uplift the ionosphere layer and support plasma bubble growth (Chou et al. 2021). Thus, it is very likely that the very-long TID wavelength, significant ionosphere uplift, and intense dynamo electric fields by the thermospheric and tropospheric Lamb waves observed in the present study result in the proponent spread-F occurrences at night on the volcano eruption day.

Summary and conclusion

In this study, multiple observation instruments in Taiwan and Japan are employed to investigate the impact of TIDs and TADs induced by the 2022 Tonga volcanic eruption. The prominent air pressure perturbations, consistent with the speed of the tropospheric Lamb wavefront, propagate at the average horizontal speed of 300.8 ± 3.9 m/s. Meanwhile, TIDs associated with thermospheric and tropospheric Lamb wavefronts, traveling at the speeds of 539.0 ± 22.3 and 338.3 ± 4.0 m/s, cause ionospheric elevations of 86–155 and 50–110 km, respectively. The precedence of TIDs over the arrivals of Lamb wavefronts by 14–39 min indicates the presence of bow waves. The ionospheric uplifts with average

rates of 21.0 ± 5.0 and 12.4 ± 3.3 m/s are estimated to correspond to 1.1 and 0.6 mV/m eastward dynamo electric fields in the ionosphere, respectively, comparable to the daily dynamo electric in the low-latitude ionosphere. These uplifts and the eastward dynamo electric fields, caused by both thermospheric and tropospheric Lamb waves, likely enhanced the growth rate of Rayleigh–Taylor instability, which resulted in occurrences of spread-F in Taiwan and Japan in the nighttime on the Tonga volcanic eruption day of 15 January 2022. In conclusion, this study examines the responses to the volcanic eruption from the Earth's surface to the ionosphere, observed at far-field distances (7800–8600 km) in Taiwan and Japan. The observed TADs, TIDs, and significant ionospheric uplifts linked to the tropospheric and thermospheric Lamb wavefronts shed light on how extreme natural events affect atmospheric and ionospheric dynamics.

Abbreviations

GNSS	Global Navigation Satellite System
INTERMAGNET	International Real-time Magnetic Observatory Network
JMA	Japan Meteorological Agency
NRLMSISE	US Naval Research Laboratory Mass Spectrometer and Incoherent Scatter Radar Extended model
ROTI	Rate of TEC index
TAD	Traveling atmospheric disturbance
TEC	Total electron content
TID	Traveling ionospheric disturbance

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40562-025-00419-0>.

Additional file 1.
Additional file 2.
Additional file 3.
Additional file 4.
Additional file 5.
Additional file 6.

Acknowledgements

The authors gratefully acknowledge the Central Weather Administration and the Marine Meteorological Center for providing the observation data in Taiwan.

Author contributions

TCL conducted the research, carried out formal analysis and visualization, and wrote the original manuscript. JYL conceptualized this study. JYL and CYL provided guidance on the research, revised the manuscript, and secured funding. PHL, THK, and YLT validated the dataset and advised on the research content and revisions, with THK also contributing to the visualization.

Funding

This study was supported by the Taiwan National Science and Technology Council under Grants NSTC 112-2123-M-008-003, NSTC 110-2111-M-008-034-MY2, and NSTC 112-2111-M-008-010. Additional support was provided by the Center for Astronautical Physics and Engineering (CAPE) through the Featured Area Research Center program, funded by the Higher Education Sprout Project of the Ministry of Education (MOE), Taiwan.

Availability of data and materials

No datasets were generated or analyzed during the current study.

Declarations**Competing interests**

The authors declare no competing interests.

Received: 29 March 2025 Accepted: 7 September 2025

Published online: 29 September 2025

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